

MASTER PROMPT: SOVEREIGN HUB DIGITAL TWIN DASHBOARD

GOAL: Generate a single, complete, and runnable **React (JSX) application** named `SovereignHubMaster.jsx`. This dashboard must integrate all persistence, aesthetic, interactive, and simulated \text{AI} features outlined below, using only standard React hooks, Tailwind CSS, and the provided Firebase/Gemini API global structures.

I. Technical & Aesthetic Foundation

1. **Technology:** Single-file React (JSX). All logic, components, state, and necessary CSS overrides must be in this file. The main component MUST be `App` and be the default export.
2. **Styling:** Use **Tailwind CSS**. The theme must be **dark, high-contrast, minimalist luxury** (deep black/indigo background, white text, metallic blue/gold accents). The design MUST be fully **responsive**, prioritizing excellent mobile layout first, and utilizing a multi-column layout for desktop.
3. **Dependencies:** Assume availability of Tailwind CSS. Icons must be provided using inline SVG or placeholders for common concepts (like wallet, health, lightning bolt).

II. Data Persistence & Authentication (Firestore)

The application MUST connect to and use the provided Canvas global variables for Firebase operations and manage all state in the database.

1. **Initialization:** Initialize Firebase services (db, auth) using the provided global variables (`__firebase_config`, `__app_id`).
2. **Authentication:** Use a `useEffect` hook to authenticate the user immediately upon load using `signInWithCustomToken(__initial_auth_token)` or falling back to `signInAnonymously()` if the token is unavailable. Derive and store the unique `userId`.
3. **Data Path:** All primary data must be stored in the private user collection path: `/artifacts/${appId}/users/${userId}/hub\_data/main\_dashboard`.
4. **State Schema:** The primary application state (stored in the Firestore document \text{ID} `main_dashboard`) must include:
 - `eVppBalance`: Number (Initial: 10.00)
 - `peaceBalance`: Number (Initial: 100.00)
 - `gearBalance`: Number (Initial: 50)
 - `lastWorkoutTimestamp`: String (ISO date/time, empty initially)
 - `prawnHealth`: Object ({ `pH`: 7.0, `oxygen`: 95 })
5. **Real-Time Sync:** Use an `onSnapshot` listener to fetch and listen for real-time changes to the `main_dashboard` document.

III. Interactive 3D Digital Twin Viewer

Implement a rotatable model using CSS 3D transformations within a dedicated component to simulate the **Sovereign Hub Digital Twin**.

1. **Interaction:** The component must allow users to **drag (mouse or touch)** to freely rotate

the model (simulated as a cube).

2. **State & Clamping:** Use `useState` to track rotation $\{x, y\}$. The X-axis rotation must be **clamped between -90° and 90°** to prevent the model from flipping.
3. **Visual:** Inject necessary CSS for perspective, `transform-style: preserve-3d`, and the specific transformations required to position the 6 cube faces correctly.
4. **Face Labels:** Label the six faces with module names (e.g., **VPP CORE**, **AQUAPONICS**, **SENSORS**, **W3 PORT**).

IV. Core Functionality Modules

The application `\text{UI}` must feature the following interactive modules, housed in distinct, well-designed panels.

1. `\text{E}_{\text{VPP}}` Minting (Simulated `\text{AI}` Call)

- **UI:** A prominent button labeled **"Start `\text{E}_{\text{VPP}}` Workout (30 Min)"**.
- **Action:** When clicked, disable the button and simulate an `\text{AI}` call using the provided fetch structure.
 - **Model:** `gemini-2.5-flash-preview-09-2025`.
 - **User Query:** "Generate a single-sentence, highly motivational summary of a 30-minute high-intensity interval training (`\text{HIIT}`) workout."
 - **Reward Logic:** Upon receiving the `\text{AI}` response, award a **random whole number of `\text{E}_{\text{VPP}}` tokens between 5 and 15**.
 - **Update:** Update `eVppBalance`, set `lastWorkoutTimestamp` to the current time, and save the updated state to Firestore.
 - **Feedback:** Display the `\text{AI}`'s motivational summary and the amount of `\text{E}_{\text{VPP}}` minted in a clear notification area.

2. Token Wallet & Exchange

- **Display:** Clearly show the three token balances: `\text{E}_{\text{VPP}}`, `\text{PEACE}`, and `\text{Gear}` Tokens (GT).
- **"Purchase Gear" Action:** Include a button next to the GT balance labeled "Purchase Gear."
 - **Action:** When clicked, deduct a static amount of **5 GT** from the `gearBalance` and save the new state to Firestore.
 - **Constraint:** Only allow the transaction if `gearBalance` is ≥ 5 .

3. Biological Health Monitor

- **Display:** Show the `\text{pH}` and `\text{O}_2` Saturation values from the `prawnHealth` state.
- **Critical Alert:** Implement dynamic styling for critical values:
 - If `\text{pH} < 6.8` or `\text{pH} > 7.5`, the `\text{pH}` display must use **red, bold text**.
 - If `\text{O}_2 \text{ Saturation} < 90`, the `\text{O}_2` display must use **red, bold text** and an alarming visual indicator.